

Presentación:

Nombre:

Algelis Severino

Tema:

Introducción a la Programación y Diagrama de flujo.

Tecnico:

Ciberseguridad Semipresencial

Facilitador:

Kevin Feliz Henriquez

Centro:

Diapiato Pro salud

Descripción:

En este documento .pdf adjunto capturas del reto 11 de 30 days of code de HackerRank.

Link GitHub:

<https://github.com/SeverinoCyber/30-days-of-code/tree/main/Hackerrank/day20>

Python3:

Problem

Submissions

Leaderboard

Discussions

Editorial

Objective

Today, we're discussing a simple sorting algorithm called Bubble Sort. Check out the [Tutorial](#) tab for learning materials and an instructional video!

Consider the following version of Bubble Sort:

```
for (int i = 0; i < n; i++) {
    // Track number of elements swapped during a single array traversal
    int numberOfSwaps = 0;

    for (int j = 0; j < n - i; j++) {
        // Swap adjacent elements if they are in decreasing order
        if (a[j] > a[j + 1]) {
            swap(a[j], a[j + 1]);
            numberOfSwaps++;
        }
    }

    // If no elements were swapped during a traversal, array is sorted
    if (numberOfSwaps == 0) {
        break;
    }
}
```

Task

Given an array, `a`, of size `n` distinct elements, sort the array in ascending order using the Bubble Sort algorithm above. Once sorted, print the following 3 lines:

1. Array is sorted in `numSwaps` swaps.
where `numSwaps` is the number of swaps that took place.

2. First Element: `firstElement`
where `firstElement` is the first element in the sorted array.

3. Last Element: `lastElement`
where `lastElement` is the last element in the sorted array.

Hint: To complete this challenge, you will need to add a variable that keeps a running tally of all swaps that occur during execution.

Example

Change Theme

Language

Python 3

Exit Full Screen View

```
1  #!/bin/python3
2
3  import math
4  import os
5  import random
6  import re
7  import sys
8
9
10
11 ✓ if __name__ == '__main__':
12     n = int(input().strip())
13
14     a = list(map(int, input().rstrip().split()))
15
16     # Write your code here
17     total_swaps = 0
18     for i in range(n):
19         numberswap = 0;
20
21         for j in range(n-1):
22             if (a[j] > a[j+1]):
23                 a[j], a[j + 1] = a[j+1], a[j]
24                 numberswap+=1
25                 total_swaps += 1
26
27         if(numberswap == 0):
28             break
29     print(f"Array is sorted in {total_swaps} swaps.")
30     print(f"First Element: {a[0]}")
31     print(f"Last Element: {a[-1]}")
32
```

Line: 32 Col: 1

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        if (a[j] > a[j + 1]) {
            swap(a[j], a[j + 1]);
            numberOfSwaps++;
        }
    }

    // If no elements were swapped during a traversal, array is sorted
    if (numberOfSwaps == 0) {
        break;
    }
}
```

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Example

32

Line: 32 Col: 1

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Congratulations

You solved this challenge. Would you like to challenge your friends?

[f](#) [t](#) [in](#)

Next Challenge

Test case 0

Compiler Message

Success

Test case 1

Input (stdin)

Download

```
1 3
2 1 2 3
```

Test case 2

Expected Output

Download

```
1 Array is sorted in 0 swaps.
2 First Element: 1
3 Last Element: 3
```

JavaScripts:

Problem

Today, we're discussing a simple sorting algorithm called Bubble Sort. Check out the [Tutorial](#) tab for learning materials and an instructional video!

Consider the following version of Bubble Sort:

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for (int i = 0; i < n; i++) {
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    int numberOfSwaps = 0;

    for (int j = 0; j < n - i; j++) {
        // Swap adjacent elements if they are in decreasing order
        if (a[j] > a[j+1]) {
            swap(a[j], a[j+1]);
            numberOfSwaps++;
        }
    }

    // If no elements were swapped during a traversal, array is sorted
    if (numberOfSwaps == 0) {
        break;
    }
}
```

Task
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Example

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```
});
function readLine() {
    return inputString[currentLine++];
}

function main() {
    const n = parseInt(readLine().trim(), 10);

    const a = readLine().replace(/\s+$/g, '').split(' ').map(aTemp => parseInt(aTemp, 10));

    // Write your code here
    let total_swaps = 0;
    for(let i = 0; i < n; i++){
        let numberswaps = 0;

        for(let j = 0; j < n - 1; j++){
            if(a[j] > a[j+1]){
                [a[j], a[j+1]] = [a[j+1], a[j]];
                total_swaps += 1;
                numberswaps += 1;
            }
        }
        if(numberswaps == 0){
            break;
        }
    }

    console.log("Array is sorted in " + total_swaps + " swaps.");
    console.log("First Element: " + a[0]);
    console.log("Last Element: " + a[a.length - 1]);
}
```

Line: 23 Col: 1

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Problem

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        if (a[j] > a[j+1]) {
            swap(a[j], a[j+1]);
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        }
    }

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        break;
    }
}
```

Task
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Example

46
47
48
49
50

```
console.log("Array is sorted in " + total_swaps + " swaps.");
console.log("First Element: " + a[0]);
console.log("Last Element: " + a[a.length - 1]);
}
```

Line: 23 Col: 1

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You solved this challenge. Would you like to challenge your friends? [f](#) [t](#) [i](#) [n](#)

Next Challenge

Test case 0

Compiler Message

Success

Test case 1

Input (stdin)

Download

1 3
2 1 2 3

Test case 2

Expected Output

Download

1 Array is sorted in 0 swaps.
2 First Element: 1
3 Last Element: 3

C#:

Problem

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HackerRank

Prepare > Tutorials > 30 Days of Code > Day 20: Sorting

```
// Track number of elements swapped during a single array traversal
int numberOfSwaps = 0;

for (int j = 0; j < n - 1; j++) {
    // Swap adjacent elements if they are in decreasing order
    if (a[j] > a[j + 1]) {
        swap(a[j], a[j + 1]);
        numberOfSwaps++;
    }
}

// If no elements were swapped during a traversal, array is sorted
if (numberOfSwaps == 0) {
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}
```

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Example

`a = [4, 3, 1, 2]`

```
original a: 4 3 1 2
round 1 a: 3 1 2 4 swaps this round: 3
round 2 a: 1 2 3 4 swaps this round: 2
round 3 a: 1 2 3 4 swaps this round: 0
```

In the first round, the 4 is swapped at each of the 3 comparisons, ending in the last position. In the second round, the 3 is swapped at 2 of the 3 comparisons. Finally, in the third round, no swaps are made on the iterations stop. The

Change Theme Language C#

```
16 class Solution
17 {
18     public static void Main(string[] args)
19     {
20         int n = Convert.ToInt32(Console.ReadLine().Trim());
21
22         List<int> a = Console.ReadLine().TrimEnd().Split(' ').ToList().Select(aTemp => Convert.ToInt32(aTemp)).ToList();
23
24         // Write your code here
25         int total_swaps = 0;
26         for (int i = 0; i < n; i++) {
27
28             int numberOfSwaps = 0;
29
30             for (int j = 0; j < n - 1; j++) {
31
32                 if (a[j] > a[j + 1]) {
33                     (a[j], a[j + 1]) = (a[j + 1], a[j]);
34                     numberOfSwaps++;
35                     total_swaps++;
36                 }
37             }
38
39             if (numberOfSwaps == 0) {
40                 break;
41             }
42         }
43
44         Console.WriteLine("Array is sorted in " + total_swaps + " swaps.");
45         Console.WriteLine("First Element: " + a[0]);
46         Console.WriteLine("Last Element: " + a[a.Count() - 1]);
47     }
48 }
49
50
```

Line: 46 Col: 51

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Problem

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Prepare > Tutorials > 30 Days of Code > Day 20: Sorting

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Line: 46 Col: 51

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Next Challenge

Test case 0

Compiler Message

Success

Test case 1

Input (stdin)

Download

1 3
2 1 2 3

Test case 2

Expected Output

Download

1 Array is sorted in 0 swaps.
2 First Element: 1
3 Last Element: 3